

The norm-subspace topology collapses the tentative AIII mobility-gap classification

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Claimed status: full negative result, subject to human review. For the authors' tentative v2 mobility-gap space in class AIII, the proposed operator-norm subspace topology has only one path component. In particular, the half-space index is not a path invariant. An explicit index-changing path exists even among dynamically localized Hamiltonians with uniformly finite spectral degeneracy.

1 The source problem

Chung and Shapiro consider one-dimensional local Hamiltonians H for which $\ker H$ is finite dimensional and $\text{sgn}(H)$ is local. They call this their tentative mobility-gap space $\mathcal{I}_{\text{mg},v2}$. Their sign-initial topology has too many components, so they propose either changing the topology—for example to the operator-norm subspace topology—or strengthening the space, and postpone both investigations [1, Definition 7.5 and Example 7.6]. The exact source passage is reproduced below.

Definition 7.5 (another tentative definition for mobility gap in $d = 1$). A material $H = H^* \in \mathcal{L}$ is mobility gapped at zero iff $\ker H$ is finite dimensional and $\text{sgn}(H) \in \mathcal{L}$. We denote the space of all such operators as $\mathcal{I}_{\text{mg},v2}$; its topology remains to be defined.

We note that in this case, $\text{sgn}(H)$ is merely a partial isometry with finite kernel (and so it is Fredholm, $\text{sgn}(H)$ having closed range) and that still $\Lambda \text{sgn}(H)$ is Fredholm.

Unfortunately, with the initial topology $\mathcal{T}_{\text{sgn}:\mathcal{I}_{\text{mg},v2} \rightarrow \mathcal{B}}$ from norm topology on $\mathcal{B}$, this definition is *still* not good enough, as the following counterexample demonstrates a deviation from the Kitaev table. As a result, to rectify this situation, one should either define another topology on $\mathcal{I}_{\text{mg},v2}$ (e.g. the subspace topology) or start with another space entirely. The first step should be to require an open interval around zero to have finite degeneracy (uniformly). Another possibility would be to place a dynamical constraint. We postpone such investigations to future work.

Example 7.6. *There exist two operators $R, S \in \mathcal{I}_{\text{mg},v2,\text{AIII}}$ which have the same Λ -index and yet there is no continuous path connecting them in the topology $\mathcal{T}_{\text{sgn}:\mathcal{I}_{\text{mg},v2} \rightarrow \mathcal{B}}$.*

Proof. Define R to be the unitary right shift operator on $\ell^2(\mathbb{Z})$ and Λ as above. Then $\text{sgn}(R) = \Lambda$.

Figure 1: The tentative space and proposed repairs in the source paper.

The norm-subspace proposal has the opposite problem from the sign-initial topology: it has too *few* components. In class AIII the entire tentative space is path connected. The proof also identifies the precise boundary. If an essential gap is imposed, the chiral block is Fredholm and integer indices survive. Without it, compact singular values suppress arbitrarily large changes in the polar part.

2 Proof intuition

An AIII Hamiltonian is determined by one off-diagonal operator A . The tentative mobility-gap assumptions say exactly that A and A^* have finite-dimensional kernels and that the polar partial isometry of A is local. First deform A to that polar part. Pair as many kernel and cokernel directions as possible by a finite-rank path. What remains is either an isometry or a coisometry.

Every isometry is a strong-operator limit of unitaries which differ from the identity by finite rank. Such a strong path need not be norm continuous, and its half-space index may jump. Now multiply on the right by a fixed positive, injective compact operator D . Strong continuity becomes norm continuity because multiplication by a compact operator upgrades strong convergence on bounded sets to norm convergence. Since D has dense range, the polar part remains the chosen isometry or unitary. Thus the path stays inside the tentative space while erasing all index information.

3 The off-diagonal model

Let $\mathcal{H}$ be a separable infinite-dimensional complex Hilbert space and let Λ be an orthogonal projection with both $\Lambda\mathcal{H}$ and $\Lambda^\perp\mathcal{H}$ infinite dimensional. Put

$$\mathcal{L}_\Lambda := \{A \in \mathcal{B}(\mathcal{H}) : [\Lambda, A] \in \mathcal{K}(\mathcal{H})\}.$$

For an operator A , write $A = U|A|$ for its polar decomposition, with $U = 0$ on $\ker A$. Define

$$\mathfrak{M}_\Lambda := \{A \in \mathcal{L}_\Lambda : \dim \ker A < \infty, \dim \ker A^* < \infty, U \in \mathcal{L}_\Lambda\},$$

with the operator-norm subspace topology.

On $\widehat{\mathcal{H}} = \mathcal{H} \oplus \mathcal{H}$, fix the chiral grading $\Gamma = \text{diag}(I, -I)$ and spatial projection $\widehat{\Lambda} = \Lambda \oplus \Lambda$. The map

$$A \longmapsto H_A := \begin{pmatrix} 0 & A^* \\ A & 0 \end{pmatrix}$$

is an isometry onto the self-adjoint operators anticommuting with Γ . Moreover,

$$\ker H_A = \ker A \oplus \ker A^*, \quad \text{sgn}(H_A) = \begin{pmatrix} 0 & U^* \\ U & 0 \end{pmatrix}.$$

Consequently $A \in \mathfrak{M}_\Lambda$ is exactly the off-diagonal-block form of the source's tentative AIII v2 conditions. The chiral bulk non-triviality condition is unaffected: all polar parts below are local Fredholm partial isometries, and finite-rank modifications preserve the essential non-triviality of the two spatial halves.

4 Two smoothing lemmas

Lemma 1 (compact smoothing of strong paths). *Let $T : [0, 1] \rightarrow \mathcal{B}(\mathcal{H})$ be strongly continuous and uniformly bounded. If D is compact, then $t \mapsto T(t)D$ is norm continuous.*

Proof. Fix t_0 and approximate D in norm by a finite-rank operator F . Strong continuity is uniform in operator norm on the finite-dimensional space $\text{Ran } F$, so $\|(T(t) - T(t_0))F\| \rightarrow 0$. If $\sup_t \|T(t)\| \leq M$, then

$$\|(T(t) - T(t_0))(D - F)\| \leq 2M\|D - F\|.$$

First choose F and then let $t \rightarrow t_0$. □

Lemma 2 (a local strong path to an isometry). *Let $V \in \mathcal{L}_\Lambda$ be an isometry with finite-dimensional cokernel. There is a strongly continuous path $W : [0, 1] \rightarrow \mathcal{B}(\mathcal{H})$ such that*

$$W(0) = I, \quad W(1) = V,$$

and, for every $t < 1$, $W(t)$ is a unitary satisfying $W(t) - I \in \mathcal{K}(\mathcal{H})$. In particular every $W(t)$ is Λ -local.

Proof. Choose finite-dimensional subspaces

$$\{0\} = E_0 \subset E_1 \subset E_2 \subset \dots$$

with dense union. For every $n \geq 1$, extend the finite-dimensional isometry $V|_{E_n} : E_n \rightarrow VE_n$ to a unitary W_n on $\mathcal{H}$ which is the identity on the orthogonal complement of $E_n + VE_n$. Thus

$$W_n|_{E_n} = V|_{E_n}, \quad W_n - I \text{ has finite rank.}$$

Set $W_0 = I$. The unitaries W_n and W_{n+1} agree with V on E_n . Therefore $W_{n+1}W_n^*$ fixes VE_n , differs from I by finite rank, and has a finite-rank self-adjoint logarithm X_n which vanishes on VE_n :

$$W_{n+1}W_n^* = e^{iX_n}.$$

Join W_n to W_{n+1} by $s \mapsto e^{isX_n}W_n$, and place these paths on consecutive intervals whose right endpoints tend to 1. Every intermediate unitary differs from I by finite rank and agrees with V on E_n .

For $x \in \mathcal{H}$, approximation by E_n gives, uniformly on the n -th tail interval,

$$\|(W(t) - V)x\| \leq 2 \operatorname{dist}(x, E_n) \longrightarrow 0.$$

Defining $W(1) = V$ therefore gives the required strongly continuous path. $\square$

5 Complete collapse in the norm-subspace topology

Theorem 3 (path connectedness). *The space $\mathfrak{M}_\Lambda$ is path connected. Equivalently, the tentative AIII v2 mobility-gap Hamiltonian space, equipped with the operator-norm subspace topology, satisfies*

$$\pi_0(\mathcal{I}_{\text{mg}, \text{v2}, \text{AIII}}) = \{*\}.$$

Proof. Choose once and for all a positive, injective compact operator $D \in \mathcal{L}_\Lambda$. For example, choose orthonormal bases separately in $\Lambda\mathcal{H}$ and $\Lambda^\perp\mathcal{H}$, let D be diagonal with strictly positive entries tending to zero, and note that $D\Lambda = \Lambda D$. We show that every $A \in \mathfrak{M}_\Lambda$ can be joined to D .

Step 1: polar deformation. Write $A = U|A|$ and put $P = U^*U = I - P_{\ker A}$. The path

$$A_s = U((1-s)|A| + sP), \quad 0 \leq s \leq 1,$$

joins A to U . On $P\mathcal{H}$, the positive factor is injective with dense range; for $s > 0$ it is bounded below. Thus every A_s has polar part U , kernel $\ker U$, and cokernel $\ker U^*$. It belongs to $\mathfrak{M}_\Lambda$ because $\mathcal{L}_\Lambda$ is a C^* -algebra and $|A|, U, P \in \mathcal{L}_\Lambda$.

Step 2: pair finite defects. Let

$$k = \dim \ker U, \quad c = \dim \ker U^*, \quad m = \min(k, c).$$

Choose m -dimensional subspaces $K_0 \subseteq \ker U$ and $C_0 \subseteq \ker U^*$, and a partial isometry $J : K_0 \rightarrow C_0$. The path

$$U_s = U + sJ, \quad 0 \leq s \leq 1,$$

lies in $\mathfrak{M}_\Lambda$. For $s > 0$, its polar part is $U + J$; all changes are finite rank. At the endpoint $V = U + J$, either

$$V^*V = I \quad (k \leq c) \quad \text{or} \quad VV^* = I \quad (c \leq k).$$

Thus V is an isometry or a coisometry (and is unitary if $k = c$).

Step 3: erase an isometry by compact attenuation. Suppose first that V is an isometry. The path

$$V((1-s)I + sD), \quad 0 \leq s \leq 1,$$

joins V to VD inside $\mathfrak{M}_\Lambda$, with constant polar part V . Let W be the strong path from Lemma 2. Lemma 1 says that

$$t \mapsto W(1-t)D$$

is a norm-continuous path from VD to D . Its polar part is $W(1-t)$: at interior parameters this is a local unitary, and at the initial endpoint it is V . The kernels and cokernels are always finite.

If V is a coisometry, apply the preceding construction to the isometry V^* and take adjoints. This again joins V to the self-adjoint operator D . Concatenating the three steps joins every A to the same base point D , proving path connectedness. $\square$

Corollary 4. *The map*

$$A \mapsto \text{ind}\left(\Lambda \text{ polar}(A) \Lambda + \Lambda^\perp\right)$$

is not continuous, and is not constant on norm paths in $\mathfrak{M}_\Lambda$.

6 An explicit dynamically localized index jump

The preceding theorem classifies the tentative deterministic space. The failure of the raw norm topology persists even if one keeps the source's stronger dynamical localization and uniformly finite spectral-degeneracy conditions.

Let $\mathcal{H} = \ell^2(\mathbb{Z})$, with basis $(\delta_n)_{n \in \mathbb{Z}}$, and let Λ project onto the span of $\{\delta_n : n \geq 1\}$. Let

$$R\delta_n = \delta_{n+1}, \quad D\delta_n = 2^{-|n|}\delta_n.$$

For $N \geq 1$, define a finite cyclic shift

$$C_N\delta_n = \begin{cases} \delta_{n+1}, & -N \leq n < N, \\ \delta_{-N}, & n = N, \\ \delta_n, & |n| > N. \end{cases}$$

Then $C_N - I$ has finite coordinate support and $C_N \rightarrow R$ strongly.

Proposition 5 (explicit index-changing mobility-gap path). *There is a strongly continuous path of local unitaries $U : [0, 1] \rightarrow \mathcal{B}(\mathcal{H})$ such that*

$$U(0) = I, \quad U(1) = R,$$

and $U(t) - I$ has finite coordinate support for every $t < 1$. Consequently

$$A_t := U(t)D, \quad H_t := \begin{pmatrix} 0 & A_t^* \\ A_t & 0 \end{pmatrix}$$

is an operator-norm-continuous path in the dynamically localized AIII mobility-gap space, while

$$\text{ind}_\Lambda(\text{polar } A_t) = \begin{cases} 0, & t < 1, \\ -1, & t = 1. \end{cases}$$

Proof. Put $t_N = 1 - 2^{-N}$. Connect I to C_1 on $[0, t_1]$. On $[t_N, t_{N+1}]$, connect C_N to C_{N+1} inside the finite-dimensional unitary group supported on $\text{span}\{\delta_n : |n| \leq N+1\}$, choosing the path to agree with R on

$$E_N := \text{span}\{\delta_n : -N \leq n < N\}.$$

This is possible because C_N and C_{N+1} already agree there. Define $U(1) = R$. Agreement on E_N proves strong continuity at 1.

More is true after attenuation. For $t \in [t_N, t_{N+1}]$,

$$(U(t) - R)DP_{E_N} = 0, \quad \|DP_{E_N^\perp}\| = 2^{-N}.$$

Hence

$$\|(U(t) - R)D\| \leq 2^{1-N} \rightarrow 0.$$

Thus $t \mapsto A_t$, and therefore $t \mapsto H_t$, is norm continuous. Since D is positive, injective, and has dense range, $\text{polar}(A_t) = U(t)$. For $t < 1$, $U(t)$ is a finite-rank perturbation of I , so its half-space index is zero. At the endpoint, the compression of R is the unilateral shift and has index -1 .

It remains to check the stronger mobility-gap conditions. Let

$$H_D = \begin{pmatrix} 0 & D \\ D & 0 \end{pmatrix}, \quad Q_t = \text{diag}(I, U(t)).$$

Then $H_t = Q_t H_D Q_t^*$, so the spectrum is independent of t : it consists of the eigenvalues $\pm 2^{-|n|}$, with uniformly bounded multiplicity, accumulating only at zero, and zero is not an eigenvalue. For $t < 1$, $Q_t - I$ has finite coordinate support. Therefore every bounded Borel function $f(H_t) = Q_t f(H_D) Q_t^*$ is position diagonal outside a finite coordinate set and satisfies the source's dynamical decay estimate (with constants depending on t). At $t = 1$, R has propagation one, so the same conclusion holds. Taking, for example, $\Delta = (-2, 2)$ verifies the mobility-gap conditions along the path. $\square$

The parenthetical qualification is essential: the raw norm-subspace topology does not require localization constants to be uniform along a path. A stronger topology controlling those constants or controlling the Fermi projection may still preserve the index; this is precisely the type of topology studied in [3].

7 Why the essential-gap result does not collapse

For comparison, let

$$\mathfrak{F}_\Lambda := \{A \in \mathcal{L}_\Lambda : A \text{ is Fredholm}\}.$$

Relative to $\mathcal{H} = \Lambda^\perp \mathcal{H} \oplus \Lambda \mathcal{H}$, locality gives

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad b, c \in \mathcal{K}.$$

Thus A is Fredholm exactly when a and d are Fredholm. Scaling the compact off-diagonal blocks to zero and using fixed-index connectivity of Fredholm operators gives

$$\pi_0(\mathfrak{F}_\Lambda) \cong \mathbb{Z}^2, \quad A \longmapsto (\text{ind } A, \text{ind } d).$$

On the ordinary-index-zero sector, the second coordinate alone yields $\mathbb{Z}$, the expected AIII entry. This is the earlier substantial partial result, now seen as the sharp boundary of Theorem 3: the compact attenuator D is not Fredholm, so the collapsing path necessarily leaves the essential-gap space.

8 Verification, novelty, and scope

Every construction above has an explicit membership check:

1. polar deformation keeps the polar part and finite defect spaces fixed;
2. defect pairing changes the polar part only by finite rank;
3. strong unitary paths consist of finite-rank perturbations of the identity before their endpoint;
4. compact attenuation converts those strong paths into norm paths;
5. the explicit path is unitarily equivalent to a fixed diagonal compact Hamiltonian, so spectral multiplicities and localization are transparent.

The companion script checks finite cyclic shifts, the exact agreement subspaces, polar identities on finite windows, and the decay of $\|(C_N - R)D\|$. It is a diagnostic, not part of the proof.

A bounded search of the run indexes and primary arXiv sources found no prior statement of this path-connectedness theorem or of the explicit index-changing path. Shapiro’s earlier topology deliberately controls mobility-gap data so that invariants remain continuous [3]. The authors’ 2026 Part III paper proves the spectral-gap classification and still postpones the mobility-gap Hamiltonian/projection connection [2, §2]. Novelty confidence is *moderate*: the compact smoothing lemma is classical, but its use here gives a direct and apparently new resolution of the source’s proposed norm topology.

The result is full for the tentative v2 AIII space with the raw norm-subspace topology. It does not classify all ten symmetry classes, nor a topology which imposes uniform localization constants or continuity of the Fermi projection.

References

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- [3] J. Shapiro, *The Topology of Mobility-Gapped Insulators*, Letters in Mathematical Physics **110** (2020), 1153–1172; arXiv:1811.07997.